The Perks of Being a First Mover

Does White's Advantage in Chess Change with Rating, Opening and Time Control?

CS215 Open Project · Final Report · Group 02

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https://github.com/cs215-cse-iitb/dai-aug-2026-openproject-group-2

Section 1: Meta Information

1. Open Project Title	The Perks of Being a First Mover: Does White's Advantage in Chess Change with Rating, Opening and Time Control?
2. Group ID	"02"
3. Mentor Name	Namish Bansal
4. Group Members and Roll Numbers	Sanidhya (25b0944); Daksh Maahor (25b0974); Sahil (25b0991); VadaPavJai (25b1030)
5. Honor Code and Discussions	None
6. LLM Usage Links	**[Readable links to all chat sessions / prompts used. If an AI agent was used, export its report as PDF, host it and link it here.]**
7. Video Presentation	**[link to video]**
8. GitHub Repository	https://github.com/cs215-cse-iitb/dai-aug-2026-openproject-group-2
9. GitHub Handles and Initial Commit	
Sanidhya	https://github.com/SS3642 earliest commit: October 21, 2026
Daksh Maahor	https://github.com/Daksh-Maahor earliest commit: November 3, 2026
Sahil	https://github.com/Sahil-Deo-IITB earliest commit: November 3, 2026
VadaPavJai	https://github.com/nauseated-potato earliest commit: October 18, 2026
10. Contribution Breakdown	Unequal

Work split (for reference).	Sanidhya	Statistical modelling: fitting h, confidence intervals and the z-tests.
	Daksh Maahor	Data cleaning and opening classification; win/draw/loss rates.
	Sahil	Data engineering: the C++ streaming parser and the Colab pipeline
	VadaPavJai	Visualisation: figures, the Plotly dashboard and the report.

Section 2: Technical Content

Core Problem Statement

We measure the first-move advantage on Lichess. Instead of raw win rates, we fit White's advantage as a number of *rating points*, h, after correcting each game for the rating gap between the players. Under the Elo model a player d points above the opponent scores $E(d) = 1/(1 + 10^{-d/400})$; we give White an extra h points, $E_i(h) = 1/(1 + 10^{-(R_i^{\mathrm{W}} + h - R_i^{\mathrm{B}})/400})$, and choose the h that makes expected scores equal actual scores ($S_i = 1, \frac{1}{2}, 0$). This is the maximum-likelihood estimate (Newton's method; standard error from the Fisher information). Overall, $h =$ **[HAll]** $\pm$ **[HAllCI]** rating points. We test three hypotheses:

- **H1 (Rating).** h peaks at intermediate ratings and shrinks at 2200+.
- **H2 (Openings).** Four aggressive 1.e4 gambits (King's, Evans, Danish, Smith-Morra) carry a larger h than four drawish openings (Petrov, Berlin, QGD, French Exchange), and the drawish ones have more draws at every rating.
- **H3 (Time control).** h grows from Bullet to Classical.

Data Source(s)

Lichess Open Database (`https://database.lichess.org`, CC0), standard rated games for **[MonthList]**: one zstd-compressed PGN per month, tens of GB each, openly downloadable. We use seven of 17 header tags: `Result`, `WhiteElo`, `BlackElo`, `Event`, `TimeControl`, `Termination`, `Opening`. We read **[GamesRead]** games and keep **[GamesUsed]** (**[PctUsed]**).

Month	Games read	Games kept	Kept
[filled in by white_advantage]			
Total	**[GamesRead]**	**[GamesUsed]**	**[PctUsed]**

Tools and Software Packages

A C++ program streams the zstd files, reads only header lines, and counts games by opening, speed, 100-point rating bracket and rating gap (two months on two threads, run on Google Colab, streaming directly from Lichess). It writes `counts.csv` (a few MB); all tables, tests, figures, this report (LaTeX with pgfplots) and an interactive Plotly dashboard (`output/final/dashboard.html`) are rebuilt from it in seconds. **[list exact Python/R packages and versions used for the analysis, if any]**

Challenges

Cleaning and design. We drop games with no result (**[SkipNoResult]**), abandoned games (**[SkipAbandoned]**), missing ratings (**[SkipNoRating]**) and rating gaps above 100 (**[SkipGap]**). Games go into 100-point brackets by average rating and into a speed from `Event`. Openings are matched by Lichess name prefix (excluding gambit side-lines inside the drawish openings) and cover **[OpeningGames]** games (**[OpeningShare]**). With millions of games every difference has a tiny p-value, so we report effect sizes with 95% CIs and treat a few rating points as small. **[technical hurdles: e.g. file size/streaming, opening-name matching, memory, runtime]**

Results Achieved

Overall. White wins **[WhiteAll]**, draws **[DrawAll]**, loses **[BlackAll]**; average score **[ScoreAll]** versus **[ExpectedAll]** expected from ratings alone, i.e. $h =$ **[HAll]** $\pm$ **[HAllCI]**.

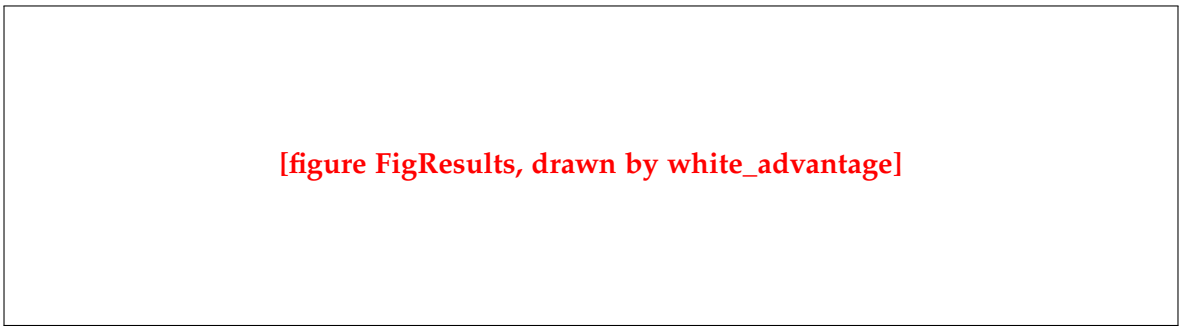


Figure 1: Share of games won by White, drawn and won by Black, by rating (all speeds).

H1: rating. We fit h separately in every 100-point rating bracket (by the average of the two ratings), for each speed, and read the shape from the chart of h against rating. We don't fit a single line through the brackets, because the shape is not close to a straight line, and a slope would hide it.

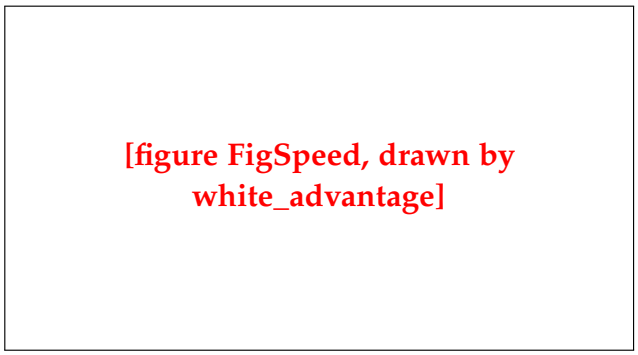


Figure 2: h per 100-point bracket, by speed (95% CIs; <1,000 games omitted).



Figure 3: h by rating: aggressive vs drawish openings.

H2: Openings. Lichess names the opening of every game, for example *King's Gambit Accepted: Bishop's Gambit*. We match those names as shown in Table 1. Inside the drawish openings we leave out their gambit side-lines, such as the Stafford Gambit in the Petrov, so that the drawish group really is quiet. The eight openings cover **[OpeningGames]** games (**[OpeningShare]** of all kept games). Everything else is "other".

Table 1: Openings and the Lichess opening names they cover. A game counts if its name starts with one of the listed names and contains none of the excluded ones.

Opening	Group	Lichess name starts with	Excluding
[filled in by white_advantage]			

Why these eight. The aggressive openings are White gambits: White gives up a pawn or more for fast development and attack (King's, Evans, Danish and Smith-Morra Gambits). The drawish openings have a reputation for symmetrical, simplified positions: the Petrov and the Berlin (Black's choice), the Queen's Gambit Declined (Black's choice), and the French Exchange (White's choice to simplify).

Each opening is fitted separately; groups are compared with a z-test, $\mathrm{SE}(\hat{h}_A - \hat{h}_D) = \sqrt{\mathrm{SE}_A^2 + \mathrm{SE}_D^2}$. Aggressive minus drawish: **[CompareDiff]** points ($z =$ **[CompareZ]**, $p =$ **[CompareP]**). **[is H2 supported?]**

Table 2: White's advantage by opening (all speeds).

Opening	Games	Avg rating	White	Draw	Black	h (pts)
[filled in by white_advantage]						
[filled in by white_advantage]						

H3: time control. Each speed is its own set of games with its own h; each speed is compared with the next faster one by the same z-test. Lichess sorts every game into a speed (UltraBullet, Bullet, Blitz, Rapid, Classical, Correspondence) by its time control. We treat each speed exactly like an opening: its games are a separate set, with their own maximum-likelihood h and win/draw/loss shares. We then compare each speed with the next faster one using the same z-test for a difference.

Table 3: White's advantage by speed (all openings).

Speed	Games	Avg rating	White	Draw	Black	h (pts)
[filled in by white_advantage]						

[figure FigOpenings, drawn by white_advantage]

Figure 4: h per opening with 95% CIs.

[figure FigSpeedBars, drawn by white_advantage]

Figure 5: h per speed with 95% CIs.

Limitations. Opening choice is not random (association, not causation); the two opening groups have different players (see Fig. 3); ratings are noisy and Lichess is not FIDE; games are not independent (active players contribute thousands), so true uncertainty is somewhat larger than our CIs; one site and two months only.

Conclusion

[answer to the three questions in a few sentences]

References

Lichess Open Database, `https://database.lichess.org`. Lichess opening names, `https://github.com/lichess-org/chess-openings`. A. E. Elo, *The Rating of Chessplayers, Past and Present*, Arco, 1978. M. E. Glickman, "Example of the Glicko-2 system", 2012.

Section 3: Frequently Answered Questions (FAQ)

Q1. How many games did you analyse and where did they come from?
[GamesUsed] standard rated Lichess games from **[MonthList]** (**[GamesRead]** read, **[PctUsed]** kept), from the CC0 Lichess Open Database.

Q2. What is h and what is its overall value?
h is the number of Elo rating points that, added to White's rating, makes the Elo-expected scores match the observed scores. Overall $h =$ **[HAll]** $\pm$ **[HAllCI]**.

Q3. Which games did you exclude and why?
Games with no result, abandoned games, missing ratings and rating gaps above 100 points: very lopsided games say little about the first move.

Q4. Why rating points instead of raw win rates?
A raw win rate mixes the first move with the rating gap and the draw rate. Rating points correct each game for the gap, so groups with different rating mixes are comparable, and draws count as half a point.

Q5. Why not fit a single slope of h against rating?
The shape is not close to a straight line, so a slope would hide it; we fit h per 100-point bracket and read the shape from the plot.

Q6. With millions of games, isn't everything "significant"?
Yes, so we report effect sizes with 95% CIs and treat differences of a few rating points as small whatever the p-value.

Q7. Which openings were compared?
Aggressive: King's, Evans, Danish and Smith-Morra Gambits. Drawish: Petrov, Berlin, Queen's Gambit Declined, French Exchange (gambit side-lines excluded).

Q8. Does the data support the hypotheses?
[H1, H2, H3 each: supported / partly / not supported, with the key number]

Q9. Does the opening *cause* the difference in h?
Not necessarily. Players choose openings, and styles, preparation and ratings differ between groups, so our results show association only.

Q10. What would you do with more time?
[e.g. more months, cluster-robust CIs by player, adjusting for player ratings, FIDE/OTB data]